

The hyperbola $ts = 1$ is a fixed-point set — but not of the involution we were looking for

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Abstract

Let $R_e(t)$ be the operator on Λ adding a connected e -ribbon with weight t^{ht} . Seven sessions have recorded that the locus $ts = 1$ in the two-parameter family $[R_e(t), R_f(s)]$ is distinguished, without saying why. The proposed explanation (Q104) was that $ts = 1$ is the fixed-point set of a bead-parity involution Θ with $\Theta R_e(t) \Theta = R_e(1/t)$.

We decide this. Three results and one honest negative.

(A) The gating computation Q107 is settled in closed form: if λ/μ is a connected e -ribbon R whose topmost row is r , then $n(\lambda) - n(\mu) = (r - 1)e + \text{maj}(R)$. The Garsia–Zabrocki exponent is therefore *never* constant in μ — not for any ribbon shape, columns included — and the conjugation is not an operator identity with a scalar. Q104 *fails as stated*. We locate the exact pin that makes the Garsia–Zabrocki theorem work: their operator adds cells at rows $1, \dots, k$ *fixed independently of μ* , whereas a ribbon floats. It is the row-pin, not the column shape.

(B) Q104 nevertheless has a true answer, and the involution is the most classical one available:

$$\omega R_e(t) \omega = t^{e-1} R_e(1/t), \quad \omega : s_\lambda \mapsto s_{\lambda'}.$$

The anomaly t^{e-1} is forced by $\text{ht}(R) + \text{ht}(R^\top) = e - 1$.

(C) Consequently $ts = 1$ *is* a fixed-point set: of the involution $\sigma(t, s) = (1/s, 1/t)$ on the parameter torus, which is implemented on operators by $\text{Ad}(\omega)$ composed with the swap of the two commutator slots. On $ts = 1$ the commutator $[R_e(t), R_e(s)]$ lies in the (-1) -eigenspace of $\text{Ad}(\omega)$.

(D) The two-bead sector of $[R_e(t), R_f(s)]$ vanishes on $ts = 1$ for *all* $e, f \geq 1$, and the mechanism is local on the Maya diagram: reversing the order of two disjoint bead moves multiplies the weight by exactly $(ts)^k$.

The negative, stated plainly. (C) does *not* imply (D). The one-bead sector also lies in the (-1) -eigenspace on $ts = 1$ and is *nonzero* there (52/52 matrix entries survive, against 0/58 in the two-bead sector). The programme’s hope that all four sightings of $ts = 1$ become corollaries of one $\mathbb{Z}/2$ is therefore **false**: two of them do, two do not.

1 Conventions and the gate

Convention 1. Λ is the ring of symmetric functions over $\mathbb{Q}(t, s)$ with Schur basis $\{s_\lambda\}$. For a partition λ , $n(\lambda) = \sum_{i \geq 1} (i - 1)\lambda_i$, $M(\lambda) = \{\lambda_j - j : j \geq 1\}$ is the Maya set, and for $e \geq 1$

$$R_e(t) s_\mu = \sum_{\lambda} t^{\text{ht}(\lambda/\mu)} s_\lambda,$$

the sum over $\lambda \supset \mu$ with λ/μ a connected border strip of size e and $\text{ht} = (\#\text{rows}) - 1$. We write $C_{e,f}(t, s) = [R_e(t), R_f(s)]$ and ω for the standard involution $s_\lambda \mapsto s_{\lambda'}$, extended $\mathbb{Q}(t, s)$ -linearly. This is Convention 2.1 of 2026-09-07-Q92-cross-rank-commutator.tex, unchanged.

Remark 2 (the gate, and it failed). The framing of Q104 carried a flagged dependency: a substitution $t = -e^\lambda$, $s = -e^\mu$ attributed to Zhang, 2603.11172, eq. (30). We opened the paper’s L^AT_EX source and located equation (30) by counting numbered environments. It is Shastry’s decorated Yang–Baxter equation for the spinful Hubbard R -matrix,

$$(\check{R}_{23}^{\uparrow\downarrow}(\lambda - \mu) + f(\lambda, \mu)\check{R}_{23}^{\uparrow\downarrow}(\lambda + \mu)\Theta_2^{\uparrow}\Theta_2^{\downarrow})\check{R}_{12}^{\uparrow\downarrow}(\lambda)\check{R}_{23}^{\uparrow\downarrow}(\mu) = \cdots,$$

in which λ, μ are *spectral parameters* and $\Theta_j = 2n_j - 1$ is a time-reversal operator on a spin-chain site. A census over the full source returns 0 hits for the exponential substitution and 0 each for “Hall–Littlewood”, “ribbon”, “Fock” and “symmetric function”. **The substitution is not in the paper.** It was a sub-agent’s construction. Zhang is dropped entirely; nothing below depends on it, and the involution used below is defined from Convention 1.

2 Q107 in closed form

Convention 3 (maj of a border strip). Let R be a connected border strip of size e occupying m rows with lengths $a_1, \dots, a_m$ read *top to bottom*. Read the cells of R starting from its bottom-left end; each successive cell is one step right or one step up. The *descent set* is $D(R) = \{p : \text{the step from cell } p \text{ to cell } p+1 \text{ is up}\} \subseteq \{1, \dots, e-1\}$, so $D(R) = \{a_m, a_m + a_{m-1}, \dots, a_m + \dots + a_2\}$ and $|D(R)| = m - 1 = \text{ht}(R)$. Put $\text{maj}(R) = \sum_{p \in D(R)} p$.

Theorem 4 (Q107). *Let $\mu \subset \lambda$ with λ/μ a connected border strip of size e and shape R , whose topmost row is r (rows indexed from 1). Then*

$$c(R, \mu) := n(\lambda) - n(\mu) = (r-1)e + \text{maj}(R).$$

Proof. Since $n(\nu) = \sum_{(i,j) \in \nu} (i-1)$, summing over cells,

$$n(\lambda) - n(\mu) = \sum_{(i,j) \in \lambda/\mu} (i-1).$$

The strip occupies rows $r, r+1, \dots, r+m-1$ with $a_1, \dots, a_m$ cells respectively, $\sum_i a_i = e$. Hence

$$n(\lambda) - n(\mu) = \sum_{i=1}^m a_i (r+i-1-1) = (r-1)e + \sum_{i=1}^m (i-1)a_i. \quad (1)$$

It remains to identify $\sigma(R) := \sum_{i=1}^m (i-1)a_i$ with $\text{maj}(R)$. Count the quantity $\Sigma = \sum_{p=1}^e \#\{d \in D(R) : d < p\}$ in two ways.

Cell-first: a cell in row i (from the top) lies in the $(m-i+1)$ -st block of the bottom-to-top reading, so exactly $m-i$ descents precede it. Thus $\Sigma = \sum_{i=1}^m a_i(m-i)$.

Descent-first: each $d \in D(R)$ is counted once for every p with $d < p \leq e$, i.e. $e-d$ times. Thus $\Sigma = \sum_{d \in D(R)} (e-d) = \text{ht}(R)e - \text{maj}(R)$, using $|D(R)| = m-1 = \text{ht}(R)$.

Since $(m-i) + (i-1) = m-1 = \text{ht}(R)$ for every i ,

$$\sum_{i=1}^m a_i(m-i) + \sum_{i=1}^m a_i(i-1) = \text{ht}(R) \sum_i a_i = \text{ht}(R)e,$$

so $\sigma(R) = \text{ht}(R)e - \Sigma = \text{ht}(R)e - (\text{ht}(R)e - \text{maj}(R)) = \text{maj}(R)$. Substituting into (1) gives the claim. $\square$

Corollary 5 (Q107 answers (b), and Q104 fails as stated). *For every $e \geq 1$ and every ribbon shape R , $c(R, \mu)$ is not constant as μ varies: its value set is $\{(r-1)e + \text{maj}(R)\}$ over the attainable top-rows r , an infinite arithmetic progression of common difference e . Its level sets are exactly the fibres of r . Consequently the Garsia–Zabrocki $t \rightarrow 1/t$ conjugation of $R_e(t)$ emits an exponent that varies with the matrix element, and there is no scalar c with $\Theta R_e(t) \Theta = t^c R_e(1/t)$ obtained by that mechanism.*

Remark 6 (where the constancy in Garsia–Zabrocki actually comes from). Garsia–Zabrocki, *Poly-nomiality of the q, t -Kostka Revisited*, [math/0008199](#), Theorem 2.3, is proved by setting $t \rightarrow 1/t$ in their (2.11) to obtain

$$t^{-n(\mu)} \mathcal{H}_k(X; 1/t) H_\mu[X; t] = t^{-n(\mu+1^k)} H_{\mu+1^k}[X; t],$$

and then, verbatim, “and this (using 2.10) may be rewritten as” $t^{\binom{k}{2}} \mathcal{H}_k(X; 1/t) H_\mu[X; t] = H_{\mu+1^k}[X; t]$.

The step “using 2.10” is exactly the assertion $n(\mu+1^k) - n(\mu) = \binom{k}{2}$, i.e. our c for the shape they add. It is constant — but the reason is *not* that 1^k is a column. The operation $\mu \mapsto \mu+1^k$ adds one cell to each of rows $1, \dots, k$, a set of rows fixed *independently* of μ ; by (1) the exponent is then $\sum_{i=1}^k (i-1) = \binom{k}{2}$ whatever μ is. A ribbon has no such pin: it floats, and r moves.

Two consequences worth recording, both of which correct the reading this session inherited.

- (i) The prediction carried in from the browse log of 2026-09-08 — that c is “constant in μ only for columns” — is **false even for columns**. A column ribbon 1^k added at top-row r has $c = (r-1)k + \binom{k}{2}$; for $e = 3$ the shape $(1, 1, 1)$ realises $c \in \{3, 6, 9, 12, \dots\}$. The prediction was shape-correct about *which theorem* was doing the work and wrong about *why*.
- (ii) $\mu \mapsto \mu+1^k$ is in general a *disconnected* skew shape, not a ribbon, and its c is constant. So the negative control proposed in the session brief — “break the ribbon condition and check c is not constant there” — tests the wrong variable. Connectedness is irrelevant; row-pinning is everything.

3 Q104: the involution exists, and it is ω

Lemma 7. *Let λ/μ be a connected border strip of size e . Then λ'/μ' is a connected border strip of size e , and*

$$\text{ht}(\lambda/\mu) + \text{ht}(\lambda'/\mu') = e - 1.$$

Proof. Transposition is an involution on Young diagrams preserving containment, so λ'/μ' is a skew shape of size e ; it is connected and contains no 2×2 block because both properties are transposition-invariant. For the height: let the strip occupy ρ rows and γ columns. Reading its e cells from one end to the other, the $e-1$ steps between consecutive cells are each either a row-change or a column-change, and the row-changes number $\rho-1$, the column-changes $\gamma-1$. Hence $(\rho-1) + (\gamma-1) = e-1$, i.e. $\rho + \gamma = e+1$. Transposition exchanges rows and columns, so $\text{ht}(\lambda'/\mu') = \gamma-1 = e-\rho = e-1 - (\rho-1) = e-1 - \text{ht}(\lambda/\mu)$. $\square$

Theorem 8 (Q104, corrected form). *For every $e \geq 1$,*

$$\omega R_e(t) \omega = t^{e-1} R_e(1/t).$$

Equivalently, $\tilde{R}_e(t) := t^{-(e-1)/2} R_e(t)$ satisfies $\omega \tilde{R}_e(t) \omega = \tilde{R}_e(1/t)$.

Proof. Both sides are $\mathbb{Q}(t)$ -linear, so it suffices to evaluate on s_μ . Since $\omega^2 = \text{id}$ and $\omega s_\nu = s_{\nu'}$,

$$\omega R_e(t) \omega s_\mu = \omega R_e(t) s_{\mu'} = \omega \sum_{\nu \supset \mu', \nu/\mu' \text{ } e\text{-strip}} t^{\text{ht}(\nu/\mu')} s_\nu = \sum_{\nu \supset \mu'} t^{\text{ht}(\nu/\mu')} s_{\nu'}.$$

As ν runs over the partitions with ν/μ' a connected e -strip, $\lambda := \nu'$ runs over the partitions with λ/μ a connected e -strip (Lemma 7 applied in both directions), and $\text{ht}(\nu/\mu') = e - 1 - \text{ht}(\lambda/\mu)$. Hence the sum equals

$$\sum_{\lambda \supset \mu} t^{e-1-\text{ht}(\lambda/\mu)} s_\lambda = t^{e-1} \sum_{\lambda \supset \mu} (1/t)^{\text{ht}(\lambda/\mu)} s_\lambda = t^{e-1} R_e(1/t) s_\mu. \quad \square$$

Remark 9 (why this was missed for seven sessions). The browse log of 2026-09-08 recorded, from Rozhkovskaya 2202.01544 (4.1)–(4.2), that $\omega \Gamma^+(u) \omega^{-1} = \Gamma^-(-u)$ fixes t , and concluded “ Θ cannot be ω ”. That conclusion is correct *about* $\Gamma^\pm$ and false about R_e . The two operators carry different t -statistics: Rozhkovskaya’s t sits in structure constants $(1-t)$, $(1-t^i)$, whereas the t in $R_e(t)$ is t^{ht} — and ht is precisely the statistic transposition complements. The instrument was reporting on a different referent.

4 The hyperbola as a fixed-point set

On the parameter torus $(\mathbb{C}^\times)^2$ put

$$\varsigma(t, s) = (s, t), \quad \iota(t, s) = (1/t, 1/s), \quad \sigma := \varsigma \circ \iota : (t, s) \mapsto (1/s, 1/t).$$

These generate a group $\cong (\mathbb{Z}/2)^2$. Note $\text{Fix}(\varsigma) = \{s = t\}$ and

$$\text{Fix}(\sigma) = \{(t, s) : s = 1/t\} = \{ts = 1\}.$$

Theorem 10. *For all $e, f \geq 1$,*

$$\omega C_{e,f}(t, s) \omega = t^{e-1} s^{f-1} C_{e,f}(1/t, 1/s) = -t^{e-1} s^{f-1} C_{f,e}(\sigma(t, s)).$$

In particular, for $e = f$ and (t, s) on the hyperbola $ts = 1$,

$$\boxed{\text{Ad}(\omega) C_{e,e}(t, 1/t) = -C_{e,e}(t, 1/t).}$$

Proof. The first equality is Theorem 8 applied to each of the four products in $C_{e,f}(t, s) = R_e(t) R_f(s) - R_f(s) R_e(t)$, using $\omega^2 = \text{id}$ to insert $\omega \omega$ between the factors. The second is $C_{e,f}(a, b) = -C_{f,e}(b, a)$, valid by definition of the commutator, applied with $(a, b) = (1/t, 1/s)$ so that $(b, a) = (1/s, 1/t) = \sigma(t, s)$. For the boxed statement put $e = f$ and $s = 1/t$: then $\sigma(t, 1/t) = (t, 1/t)$ because $ts = 1$ is the fixed locus of σ , and $t^{e-1} s^{e-1} = (ts)^{e-1} = 1$. $\square$

So $ts = 1$ is the fixed-point set of a $\mathbb{Z}/2$ action — of σ on the parameter torus, implemented on operators by $\text{Ad}(\omega)$ together with the swap of the two commutator slots. Off the hyperbola, ς and ι are distinct maps; on it they agree. That is the sense, and the only sense, in which the locus is the fixed-point set of an involution on these operators.

5 Why the two-bead sector dies, and why the $\mathbb{Z}/2$ is not the reason

Recall Theorem 4.1(2) of 2026-09-07-c2-q96-order-over-sublattice.tex: for $|M \triangle M'| = 4$, writing $M \setminus M' = \{b, c\}$, a legal assignment is an ordered pair with $b + e \in M' \setminus M$, $c + f \in M' \setminus M$ and $b, c, b + e, c + f$ pairwise distinct; with $P = \#(M \cap (b, b + e))$, $Q = \#(M \cap (c, c + f))$ and

$$k = \mathbf{1}[b + e \in (c, c + f)] - \mathbf{1}[b \in (c, c + f)],$$

one has $\langle M' | C_{e,f}(t, s) | M \rangle = \sum_{(b,c) \text{ legal}} t^{P-k} s^Q (1 - (ts)^k)$.

Theorem 11. *For all $e, f \geq 1$ the two-bead part of $C_{e,f}(t, s)$ vanishes identically on $ts = 1$. Moreover, for $e = f$ the two-assignment matrix element factors as*

$$\langle M' | C_{e,e}(t, s) | M \rangle = (1 - (ts)^k) (t^{P-k} s^Q - t^Q s^{P-k}).$$

Proof. Each summand carries the factor $1 - (ts)^k$, which vanishes when $ts = 1$ (for any $k \in \mathbb{Z}$); the first claim follows termwise. For the factorisation, $e = f$ gives exactly two legal assignments, (b, c) and (c, b) , with indices k and $-k$ and with (P, Q) exchanged. Writing $\alpha = P - k$, $\beta = Q$, the sum is

$$t^\alpha s^\beta (1 - (ts)^k) + t^{\beta+k} s^{\alpha+k} (1 - (ts)^{-k}) = (t^\alpha s^\beta - t^\beta s^\alpha) - (ts)^k (t^\alpha s^\beta - t^\beta s^\alpha),$$

since $t^{\beta+k} s^{\alpha+k} (1 - (ts)^{-k}) = t^{\beta+k} s^{\alpha+k} - t^\beta s^\alpha$ and $t^\alpha s^\beta (ts)^k = t^{\alpha+k} s^{\beta+k}$. Collecting gives the boxed form. $\square$

The factorisation makes the two exceptional loci visible as the two factors: $1 - (ts)^k$ is anti-invariant for σ and cuts out $ts = 1$; the alternant $t^\alpha s^\beta - t^\beta s^\alpha$ is anti-invariant for ς and cuts out $s = t$. It also reproves the annotation recorded on 8 September: the element vanishes identically iff $k = 0$ or $\alpha = \beta$, i.e. iff $Q = P - k$.

Proposition 12 (the mechanism, local on the Maya diagram). *Let $b \rightarrow b + e$ and $c \rightarrow c + f$ be two bead moves on a Maya set M with $b, c, b + e, c + f$ pairwise distinct and $b + e, c + f \notin M$. Let $h_e^0 = \#(M \cap (b, b + e))$ and $h_f^0 = \#(M \cap (c, c + f))$ be the heights in M , and let $h_e^{\text{after}}, h_f^{\text{after}}$ be the heights of each move performed after the other. Then*

$$h_e^{\text{after}} - h_e^0 = -k, \quad h_f^{\text{after}} - h_f^0 = +k.$$

Consequently the two orders carry weights $t^{h_e^0} s^{h_f^0 + k}$ (do e first) and $t^{h_e^0 - k} s^{h_f^0}$ (do f first), whose ratio is exactly $(ts)^k$; the two orders are assigned equal weight precisely when $ts = 1$, and the commutator then cancels in pairs.

Remark 13 (a gloss of mine that was wrong, and the computation that caught it). On first writing I asserted that reversing the order raises *both* heights by k . That is false: the shifts are $-k$ and $+k$. The conclusion survives — indeed the ratio is still $(ts)^k$ — but for a different reason: each order attributes the two heights to different variables, so the two $\pm k$ shifts appear in the ratio with the *same* sign. A sweep of 30349 disjoint move-pairs returned 4343 failures of the first statement and 0 of the corrected one.

The negative: the $\mathbb{Z}/2$ does not force the vanishing

It is tempting to read Theorem 10 as the cause of Theorem 11. It is not. Membership of the (-1) -eigenspace of $\text{Ad}(\omega)$ is not a vanishing statement, and there is a live witness to the difference *inside the same operator*: the one-bead sector of $C_{e,f}$ satisfies the same eigenvalue equation on $ts = 1$ and is nonzero there. Concretely, over $N \leq 4$:

(e, f)	one-bead entries surviving on $ts = 1$	two-bead entries surviving
$(2, 3)$	34/34	0/22
$(3, 3)$	36/36	0/6
$(3, 4)$	52/52	0/58

The reason is visible in Proposition 12: in the one-bead sector the two routes move the *same* bead ($b \rightarrow b + e \rightarrow b + e + f$ versus $b \rightarrow b + f \rightarrow b + e + f$), so they are not a reordering of one pair of moves, and their weight ratio is not of the form $(ts)^k$. Order-reversal is available only when there are two beads to reorder.

Hence the programme's stated hope — that Q104 would make all four sightings of $ts = 1$ corollaries of one involution — is refuted. What survives is: (B) and (C) explain why the locus is *distinguished*; (D) explains the two-bead vanishing, by a pairing on routes rather than a symmetry of operators. The one-bead order-of- vanishing results (Q105 Thms A and B) are *not* consequences of the $\mathbb{Z}/2$ and remain to be explained.

6 Verification

All computations are in `scratch/0909/`. Two engines were used: an abacus/Maya engine written this session, and the direct Young-diagram border-strip engine `reviews/2026-09-08-selfreview-code/ribbon.p` which is code-disjoint from it and was previously cross-checked 764/764 against Q105 Thm A.

1. **Theorem 4.** 1422 pairs (μ, strip) for $e \in \{3, 4, 5\}$, $|\mu| \leq 9$: 0 mismatches against $(r - 1)e + \text{maj}(R)$. All 28 ribbon shapes have non-constant c ; all 28 have $c - (r - 1)e$ constant.
2. **Lemma 7.** 2670 pairs, $e \leq 6$, $|\lambda| \leq 9$: 0 failures; every pair $(h, e - 1 - h)$ is realised.
3. **Theorem 8.** Verified as a matrix identity over $\mathbb{Q}(t)$ for $e = 1, \dots, 5$ and $N = 0, \dots, 8$ (up to 101×22). The two-parameter form of Theorem 10 verified for $(e, f) \in \{(2, 2), (2, 3), (3, 3), (2, 4), (3, 4)\}$, $N \leq 4$.
4. **Theorem 10, boxed form.** $\text{Ad}(\omega)C_{e,e} = -C_{e,e}$ on $ts = 1$ verified for $e = 2, 3, 4$, $N \leq 3$.
5. **Theorem 11.** Factorisation verified symbolically for $P, Q \in \{0, \dots, 5\}$, $k \in \{-2, \dots, 2\}$. Two-bead vanishing on $ts = 1$ verified for $(e, f) \in \{(1, 2), (2, 2), (2, 3), (3, 3), (3, 4), (2, 5), (4, 4)\}$: 0 survivors from 131 nonzero entries.
6. **Proposition 12.** 30349 randomly sampled disjoint move-pairs, $e, f \leq 5$: 0 failures.
7. **Controls.** (a) $n(\mu + 1^k) - n(\mu)$ is constant $= \binom{k}{2}$ for $k = 2, \dots, 5$ over all $|\mu| \leq 10$ — reproducing Garsia–Zabrocki's step “using 2.10”. (b) $\text{maj}(1^k) = \binom{k}{2}$, so Theorem 4 at $r = 1$ recovers it. (c) The same column shape added at $r \geq 2$ gives $c \in \{6, 9, 12, \dots\}$: not constant. (d) The one-bead negative control *moves* (52/52 against 0/58), so the two-bead statement is not degenerate.

7 Gaps and what is not claimed

1. Theorem 8 is proved for the operators $R_e(t)$ of Convention 1 only. No claim is made about Jing's Hall–Littlewood vertex operators $\Gamma^\pm(u)$, for which ω -conjugation is t -blind (Rozhkovskaya (4.1)–(4.2)); the two families are not identified here.
2. The scalar t^{e-1} means $\text{Ad}(\omega)$ implements $t \mapsto 1/t$ only projectively. The normalisation $\tilde{R}_e = t^{-(e-1)/2} R_e$ removes it at the cost of leaving $\mathbb{Q}(t)$ for $\mathbb{Q}(t^{1/2})$. Whether a $\mathbb{Q}(t)$ -rational normalisation exists is open and not addressed.
3. **The main gap.** Theorems 10 and 11 are both true and are *not* shown to be related. I do not have a derivation of the two-bead vanishing from the $\mathbb{Z}/2$, and the one-bead counterexample shows no such derivation can be purely eigenvalue-theoretic. A derivation would have to use the bead-number grading, and I have not found one.
4. The Q105 order-of-vanishing statements at $t = -1$ (one-bead order exactly 1; vertex-side order $\equiv N \bmod 2$) are untouched by this paper. They concern $t = -1$, a point of $\text{Fix}(\iota)$, not of $\text{Fix}(\sigma)$; whether the two loci interact is not investigated here.
5. Theorem 4 assumes λ/μ connected. The formula $(1) - n(\lambda) - n(\mu) = \sum_{\text{cells}} (\text{row} - 1)$ — holds for any skew shape; only the identification of the second term with maj uses connectedness.